

Gábor Hosu

contact@hgabor.com | +40 742 526 970 | hgabor.com | github.com/gabor-hosu | linkedin.com/in/gabor-hosu

Mathematics and computer science background with hands-on experience in NLP, GPU programming, and ML research. I build AI systems from first principles, bridging research and production through clean, reproducible implementations.

Honours and Publications

- | | |
|---|----------|
| Á. Kránics, G. Hosu , et al., "Multimodal RAG for Context-Aware Chatbots",
IEEE SISY 23rd [DOI], [Preprint] | Oct 2025 |
| 2nd Prize at 28th Transylvanian Students' Scientific Conference | May 2025 |

Skills

Languages: Python, SQL (MS SQL)

ML/AI: PyTorch, Transformers, LangChain, Sentence-Transformers, Hugging Face

Data: NumPy, Pandas, Milvus **Backend/Infra:** FastAPI, SQLAlchemy, Docker, Azure, GitLab CI/CD

Other: Git, Linux **Familiar with:** OpenCL, Java, Spring, React, C++

Experience

- | | |
|--|----------------------|
| Software Engineering Internship Codespring <ul style="list-style-type: none">Designed backend for a [RAG chatbot]: document ingestion, embedding, and retrieval pipeline.Built and deployed REST APIs (auth, file management) via Docker and GitLab CI/CD to Azure. | Jul 2024 - Sept 2024 |
| Fullstack mentorprogram Codespring
Pre-internship training program covering fullstack development with Java, Spring, and React. | Oct 2023 - May 2024 |

Projects

- | | |
|--|---------------------|
| Legal Document Form-Filling Prediction <ul style="list-style-type: none">Designed a lightweight ML model for legal document field-type prediction using text and layout context, targeting low-data settings where OCR-based approaches fail.Built and annotated a dataset of 198 legal PDFs (19 classes); achieved 75% macro F1 (5-fold CV) with a hierarchical coarse-to-fine classification approach. | Apr 2026 - May 2026 |
| Height Map Estimation From Normal Maps <ul style="list-style-type: none">Built GPU-based Poisson surface reconstruction pipeline from normal maps; derived and implemented finite-difference and FFT solvers from scratch.Benchmarked iterative (Jacobi, Gauss-Seidel) and spectral methods in OpenCL + Python with Docker; evaluated scalability and accuracy on large-scale image grids. | Mar 2026 - May 2026 |
| Embedding-Based Retrieval Paper Reproduction
[Portfolio] [GitHub] [Paper] [HF (Models & Data)] <ul style="list-style-type: none">Reproduced a 2025 Google DeepMind paper on geometric limits of single-vector embedding retrieval under GPU constraints.Built and evaluated synthetic IR pipelines (2K–50K docs), benchmarked quantized LLMs, tuned hyperparameters; presented findings at a research group seminar. | Nov 2025 - Jan 2026 |

Education

- | | |
|---|--------------------|
| Babeş-Bolyai University , MSc, Data Analysis and Modelling
Relevant coursework: NLP, Information Retrieval, Machine Learning, GPU Programming, Swarm Intelligence, Robotics | Oct 2025 - Present |
| Babeş-Bolyai University , BSc, Mathematics and Computer Science
GPA: 9.58/10 Thesis: "A file-centric chatbot based on the RAG method" | 2022 - 2025 |
- Languages:** English (Upper-Intermediate) Romanian (Intermediate) Hungarian (Native)